Roll No. : ....

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B. Tech CSE** Semester: V

Name of the Course: Introduction to Artificial Intelligence & Data Science Course Code: TCS 562

Time: 90 Minutes **Maximum Marks: 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1 (10 Marks) CO2

(a) Missionaries and Cannibals is a problem in which 3 missionaries, and 3 cannibals want to cross from the left bank of a river to the right bank of the river. There is a boat on the left bank, but it only carries at most two people at a time (and can never cross with zero people). If cannibals ever outnumber missionaries on either bank, the cannibals will eat the missionaries.

Solve the problem by showing all state space representation and draw the state space graph.

OR

(b) Consider 8-puzzle problem with following initial and goal state: CO1

Initial State

1	2	3
	4	6
7	5	8

Goal State

1	2	3
4	5	6
7	8	

Solve this 8-puzzle problem using suitable heuristic function. State the path cost value returned by the solution.

Q2 (10 Marks)

(a) Elaborate 5-number summary method. State the use of 5- number summary method in data science. Apply this method on following data and specify result achieved. CO2

26	37	24	28	35	22	31	53	41	64	29

Also, represent visual representation of resulted data.

OR

(b) A car manufacturer claims that the average miles per gallon (mpg) for their new line is less than 45 mpg. A random sample of 36 cars in this new line are tested, and the average was 43 mpg. Assume the standard deviation of all cars from this manufacturer is 5 mpg. CO2

i) State Null and Alternative hypothesis.
ii) At a 95% Confidence Interval (CI), is there enough evidence to reject the Null hypothesis.

(Use Statistic Tabular value = 1.96)

Q3	(10 Marks)	
(a)	Differentiate between supervised learning and unsupervised learning.	CO3
	OR	
(b)	Elaborate the term Skewness and Kurtosis. Discuss all types of curves under both cases.	CO2
Q4	(10 Marks)	
(a)	Explain why there is need for Exploratory data analysis (EDA) in data science. What are the necessary python libraries to perform the EDA. Discuss various feature selection techniques in EDA.	CO2
	OR	
(b)	Elaborate the term Data Science. What are the different types of data analytics. Discuss some of the applications of data science.	CO1
Q5	(10 Marks)	
(a)	A company tests three different marketing campaigns (A, B, and C) to see if they have a significant difference in average weekly sales. Campaign A had sales of {5, 7, 8, 2, 1, 3, 4} Campaign B had sales of {8, 2, 3, 6, 2, 2, 1} and Campaign C had sales of {3, 5, 8, 2, 2, 4, 7} . Calculate the F-statistics and determine if there is a significant difference in mean sales across the three campaigns at a 0.05 significance level. (Given F-statistics tabular value is 3.55)	CO6
	OR	
(b)	The sales of a company (in million dollars) for each year are shown in the table below. (see table below) i) Find the least square regression line y = a x + b. ii) Use the least squares regression line as a model to estimate the sales of the company in 2012.	CO6

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45